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System and method for vehicle bluetooth pairing

Abstract

A system for vehicle Bluetooth pairing is provided. The system includes a smartphone configured for Bluetooth communication. The system further includes a vehicle including a computerized pairing operation controller configured for Bluetooth communication. The computerized pairing operation controller includes programming to emit a notification signal detectable by the smartphone. The smartphone includes programming to establish proximity to the vehicle based upon the notification signal and establish a Bluetooth low energy connection to the vehicle based upon establishing the proximity. The computerizing pairing operation controller further includes programming to automatically set up a Bluetooth connection after the Bluetooth low energy connection is established.

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Background/Summary

INTRODUCTION

(1) The disclosure generally relates to a system and method for vehicle Bluetooth pairing.

(2) Wireless communication enables a first electronic device to communicate with a second electronic device. One form of wireless communication is Bluetooth communication, a short-range communication format. In one exemplary embodiment, Bluetooth communication may enable a smartphone to communication with a telematics module of a vehicle. Bluetooth communications may be described as a communication or wireless protocol for exchanging data over short distances, for example, up to ten meters, from fixed and mobile devices and may be useful for creating personal area networks. Bluetooth communications may use short-wavelength ultra-high frequency radio waves of a frequency range between 2.4 and 2.485 Gigahertz (GHz).

(3) Establishing communications between a first electronic device and a second electronic device may be described as pairing the devices.

SUMMARY

(4) A system for vehicle Bluetooth pairing is provided. The system includes a smartphone configured for Bluetooth communication. The system further includes a vehicle including a computerized pairing operation controller configured for Bluetooth communication. The computerized pairing operation controller includes programming to emit a notification signal detectable by the smartphone. The smartphone includes programming to establish proximity to the vehicle based upon the notification signal and establish a Bluetooth low energy connection to the vehicle based upon establishing the proximity. The computerizing pairing operation controller further includes programming to automatically set up a Bluetooth connection after the Bluetooth low energy connection is established.

(5) In some embodiments, the smartphone further includes programming to utilize an authentication certificate to provide an identity to the vehicle. The computerized pairing operation controller further includes programming to utilize a verification certificate to verify the identity of the smartphone. The programming to automatically set up the Bluetooth connection is performed after the identity is verified.

(6) In some embodiments, the notification signal includes data configured as an identifier specific to the vehicle, and the programming to establish proximity to the vehicle based upon the notification signal establishes proximity to the vehicle based upon the data configured as the identifier specific to the vehicle.

(7) In some embodiments, the data configured as an identifier specific to the vehicle is based upon a vehicle identification number for the vehicle.

(8) In some embodiments, the programming to automatically set up the Bluetooth connection includes the smartphone and the vehicle verifying identities of each other based upon a first certificate saved in the smartphone and a second certificate saved in the vehicle. The programming to automatically set up the Bluetooth connection further includes the smartphone requesting a Bluetooth name from the vehicle and the vehicle providing the Bluetooth name to the smartphone. The programming to automatically set up the Bluetooth connection further includes the smartphone requesting a pairing code from the vehicle and the vehicle providing the pairing code to the smartphone.

(9) In some embodiments, the programming to automatically set up the Bluetooth connection further includes the pairing code being displayed upon a display of the vehicle, the pairing code being displayed upon the smartphone, and completing the Bluetooth connection based upon a user providing confirmation of matching pairing codes.

(10) In some embodiments, the programming to establish the Bluetooth low energy connection to

the vehicle includes deriving a Bluetooth low energy service identifier based upon a vehicle identification number for the vehicle.

(11) In some embodiments, the programming to automatically set up the Bluetooth connection includes monitoring a single touch input by a user to the smartphone and completing the Bluetooth connection based upon the single touch input.

(12) According to one alternative embodiment, a system for vehicle Bluetooth pairing is provided. The system includes a smartphone configured for Bluetooth communication. The system further includes a vehicle including a computerized pairing operation controller configured for Bluetooth communication. The computerized pairing operation controller includes a vehicle onboarding application including programming to emit a notification signal detectable by the smartphone. The smartphone includes a smartphone onboarding application including programming to establish proximity to the vehicle based upon the notification signal and establish a Bluetooth low energy connection to the vehicle based upon establishing the proximity. The vehicle onboarding application further includes programming to automatically set up a Bluetooth connection after the Bluetooth low energy connection is established.

(13) In some embodiments, the programming to automatically set up the Bluetooth connection includes the smartphone and the vehicle verifying identities of each other based upon a first certificate saved in the smartphone and a second certificate saved in the vehicle. The programming to automatically set up the Bluetooth connection further includes the smartphone requesting a Bluetooth name from the vehicle and the vehicle providing the Bluetooth name to the smartphone. The programming to automatically set up the Bluetooth connection further includes the smartphone requesting a pairing code from the vehicle and the vehicle providing the pairing code to the smartphone.

(14) In some embodiments, the vehicle onboarding application and the smartphone onboarding application are configured for completing the Bluetooth connection based upon a single touch input by a user to the smartphone.

(15) According to one alternative embodiment, a method for vehicle Bluetooth pairing is provided. The method includes, within a first computerized processor of a vehicle, activating Bluetooth signal including a notification signal visible to Bluetooth enabled devices. The method further includes, within a second computerized processor of a smartphone, receiving the notification signal, establishing proximity to the vehicle based upon the notification signal, and establishing a Bluetooth low energy connection to the vehicle based upon establishing the proximity. The method further includes, within the first computerized processor, automatically setting up a Bluetooth connection after the Bluetooth low energy connection is established.

(16) In some embodiments, the method further includes, within the second computerized processor, utilizing an authentication certificate to provide an identity to the vehicle. The method further includes, within the first computerized processor, utilizing a verification certificate to verify the identity of the smartphone. Automatically setting up the Bluetooth connection is performed after the identity is verified.

(17) In some embodiments, the notification signal includes data configured as an identifier specific to the vehicle. Establishing proximity to the vehicle based upon the notification signal includes establishing the proximity to the vehicle based upon the data configured as the identifier specific to the vehicle.

(18) In some embodiments, the data configured as an identifier specific to the vehicle is based upon a vehicle identification number for the vehicle.

(19) In some embodiments, automatically setting up the Bluetooth connection includes verifying identities of the smartphone and the vehicle based upon a first certificate saved in the smartphone and a second certificate saved in the vehicle. Automatically setting up the Bluetooth connection further includes the smartphone requesting a Bluetooth name from the vehicle and the vehicle providing the Bluetooth name to the smartphone. Automatically setting up the Bluetooth

connection further includes the smartphone requesting a pairing code from the vehicle and the vehicle providing the pairing code to the smartphone.
(20) In some embodiments, automatically setting up the Bluetooth connection further includes the pairing code being displayed upon a display of the vehicle, the pairing code being displayed upon the smartphone, the smartphone prompting a user to provide confirmation of matching pairing codes, and completing the Bluetooth connection based upon the confirmation.
(21) In some embodiments, establishing the Bluetooth low energy connection to the vehicle includes deriving a Bluetooth low energy service identifier based upon a vehicle identification number for the vehicle.
(22) In some embodiments, automatically setting up the Bluetooth connection includes monitoring a single touch input by a user to the smartphone and completing the Bluetooth connection based upon the single touch input.
(23) The above features and advantages and other features and advantages of the present disclosure are readily apparent from the following detailed description of the best modes for carrying out the disclosure when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** illustrates a system for vehicle Bluetooth pairing, in accordance with the present disclosure;
(2) FIG. **2** schematically illustrates the smartphone of FIG. **1**, in accordance with the present disclosure;
(3) FIG. **3** schematically illustrates the computerized pairing operation controller of FIG. **1**, in accordance with the present disclosure;
(4) FIG. **4** is a flowchart illustrating a method for vehicle Bluetooth pairing, in accordance with the present disclosure;
(5) FIG. **5** is a flowchart illustrating a method for initiating a vehicle Bluetooth pairing operation, in accordance with the present disclosure;
(6) FIG. **6** is a dataflow illustrating communication between a user, an onboarding application, and a remote computer resource for a purpose of establishing simplified Bluetooth pairing, in accordance with the present disclosure;
(7) FIG. **7** is a dataflow illustrating communication between a user, a smartphone onboarding application, and a vehicle for Bluetooth pairing process, in accordance with the present disclosure;
(8) FIG. **8** is a flowchart illustrating method for operating a simplified vehicle Bluetooth pairing algorithm, in accordance with the present disclosure; and
(9) FIG. **9** is a flowchart illustrating a method for a customer to pair a Bluetooth enabled device with a vehicle including Bluetooth communications, in accordance with the present disclosure.
DETAILED DESCRIPTION
(10) Pairing a first electronic device to a second electronic device in a Bluetooth pairing operation includes establishing identities of the first device and the second device and then establishing a secure Bluetooth connection between the two devices. When a user's smartphone is paired with the electronics of a vehicle, the user may be prompted to provide information or otherwise provide inputs to one of the smartphone and/or a vehicle input device to complete the Bluetooth pairing operation. Difficult or involved pairing operations may be a dissatisfying factor for users.
(11) A system and method are provided for vehicle Bluetooth pairing. An onboarding application is setup within the electronics of a vehicle. A complimentary onboarding application is setup within a user's smartphone. Proximity of the vehicle to the user's smartphone is established. Based upon the established proximity, an automated Bluetooth pairing operation is executed.

(12) Establishing proximity of the vehicle to the user's smartphone may be performed in various ways. In one example, the smartphone may identify the vehicle based upon an identifier specific to the vehicle in the broadcast Bluetooth signal. The identifier specific to the vehicle may include or be derived from a vehicle identification number (VIN) for the vehicle. When the onboarding application in the smartphone identifies the Bluetooth signal with the given identifier, the smartphone may determine that the vehicle is in range to pair the Bluetooth. In another example, one of the electronics of the vehicle and the smartphone may broadcast a Bluetooth signal that other Bluetooth enabled devices may detect, and proximity of the vehicle to the smartphone may be established by evaluating a strength of that broadcast Bluetooth signal. For example, the strength of the broadcast Bluetooth signal may be compared to a preselected threshold signal strength, and proximity may be said to be established when the strength of the broadcast Bluetooth signal exceeds the threshold signal strength. The broadcast Bluetooth signal may be described as a notification signal or as a beacon. In another example, a location of the smartphone may be determined, a location of the vehicle may be determined, and a distance between the vehicle and the user's smartphone may be calculated. The distance may be compared to a preselected threshold distance, and when the distance is less than the threshold distance, the proximity of the vehicle to the smartphone may be said to be established.
(13) Referring now to the drawings, wherein like reference numbers refer to like features throughout the several views, FIG. **1** illustrates a system **100** for vehicle Bluetooth pairing. The system **100** is illustrated including a vehicle **120**. The vehicle **120** is illustrated including a cellular wireless communications module **122**. The vehicle is further illustrated including a computerized pairing operation controller **130**. The computerized pairing operation controller **130** is illustrated in electronic communication with a Bluetooth communications module **132** and with a touchscreen input/output device **134**. The computerized pairing operation controller **130** may additionally include internal electronics for monitoring a global positioning service (GPS) location.
(14) The system **100** is further illustrated including a smartphone device **110** which includes an input/output screen **112**. The smartphone device **110** includes internal electronics configured to communicate wirelessly over a cellular communications network and through Bluetooth communications.
(15) The system **100** is further illustrated including a cellular tower **142** configured for wireless cellular communications with the vehicle **120** and with the smartphone **110**. A remote server device **140** is illustrated in electronic communication with the cellular tower **142** and is configured for communication and data transfer with and between the vehicle **120** and the smartphone **110**.
(16) The smartphone **110** and the electronics of the vehicle **120** are each configured for Bluetooth communication, and each may broadcast a notification signal or a beacon detectable by other nearby electronic devices configured for Bluetooth communication. The smartphone **110** and the electronics of the vehicle **120** are each configured for determining a location of the respective device. The location may be determinable by GPS signal, cellular tower signal triangulation, or other similar methods.
(17) The computerized pairing operation controller **130** may include a vehicle onboarding application or a computerized program configured for operating the system and method disclosed herein. The computerized pairing operation controller **130** may include a Bluetooth application programming interface (API) which may provide functionality to enable a server-client relationship between the vehicle **120** as the server and the smartphone **110** as the client.
(18) The smartphone **110** may include a smartphone onboarding application or a computerized program configured for operating the system and method disclosed herein. The smartphone **110** may include programming enabling the smartphone **110** to act as the client through the Bluetooth (API) established within the computerized pairing operation controller **130**. The smartphone **110** may, as part of setting up the smartphone onboarding application upon the smartphone **110**, include data configured as an authentication certificate useful to establish or confirm an identity of the

smartphone **110** as a valid, trusted device with which to be communicated. The computerized pairing operation controller **130** may include data configured as a validation certificate useful to confirm the identity of the smartphone **110** based upon the smartphone **110** communicating data based upon its authentication certificate.

(19) In one embodiment, a pairing operation between the vehicle **120** and the smartphone **110** may include three steps. A first step may include the vehicle providing a notification signal or a beacon which notifies the smartphone **110** of proximity of the vehicle **120** to the smartphone **110**. A second step may include the smartphone onboarding application executed by the smartphone **110** establishing a secure Bluetooth low energy (BLE) connection between the smartphone **110** and the computerized pairing operation controller **130** of the vehicle **120**. A third step may include setting up a classic Bluetooth connection between the smartphone **110** and the vehicle **120** via automation. Throughout the disclosure, a classic Bluetooth connection may be described as Bluetooth Basic Rate/Enhanced Data Rate (BR/EDR). The operations performed by each of the smartphone **110** and the vehicle **120** may be reversed in alternative embodiments.

(20) The remote server device **140** may provide the vehicle onboarding application and the smartphone onboarding application for download. The remote server device **140** may facilitate communication between the smartphone **110** and the vehicle **120**, for example, providing location data from one device to the other device.

(21) FIG. **2** schematically illustrates the smartphone **110** of FIG. **1**. The smartphone **110** includes a computerized processing device **210**, a communications device **220**, an input/output coordination device **230**, and a memory storage device **240**. It is noted that the smartphone **110** may include other components and some of the components are not present in some embodiments.

(22) The processing device **210** may include memory, e.g., read-only memory (ROM) and random-access memory (RAM), storing processor-executable instructions and one or more processors that execute the processor-executable instructions. In embodiments where the processing device **210** includes two or more processors, the processors may operate in a parallel or distributed manner. The processing device **210** may execute the operating system of the smartphone **110**. Processing device **210** may include one or more modules executing programmed code or computerized processes or methods including executable steps. Illustrated modules may include a single physical device or functionality spanning multiple physical devices. The processing device **210** may further include programming modules, including a phone operation module **212**, a smartphone onboarding module **214**, and a Bluetooth programming module **216**.

(23) The communications device **220** may include a communications/data connection with a bus device configured to transfer data to different components of the system and may include one or more wireless transceivers for performing wireless communication.

(24) The input/output coordination device **230** includes hardware and/or software configured to enable the processing device **210** to receive and/or exchange data with on-board resources of the smartphone **110**, including the input/output screen **112** of FIG. **1**, a microphone, a speaker, and buttons installed to the smartphone **110**.

(25) The memory storage device **240** is a device that stores data generated or received by the smartphone **110**. The memory storage device **240** may include, but is not limited to, flash or solid-state memory.

(26) The smartphone operation module **212** includes programming to enable smartphone **110** to operate, may include an operating system for the smartphone **110**, and may include installed applications providing configurable functionality of the smartphone **110**.

(27) The smartphone onboarding application module **214** includes programming to operate the disclosed system and method. The smartphone onboarding application module **214** may include programming to determine proximity of the vehicle **120** of FIG. **1** to the smartphone **110**. The smartphone onboarding application module **214** may include programming to utilize the authentication certificate to establish a trusted identity with the vehicle **120**. The smartphone

onboarding application module **214** may include programming to establish a BLE connection with the vehicle **120** and to establish a classic Bluetooth connection with the vehicle **120**.

(28) The Bluetooth programming module **216** includes programming to enable communication through the established Bluetooth connection with the vehicle **120**.

(29) The smartphone **110** is provided as an exemplary computerized device capable of executing programmed code to operate the disclosed system and method. A number of different embodiments of the smartphone **110** and modules operable therein are envisioned, and the disclosure is not intended to be limited to examples provided herein.

(30) FIG. **3** schematically illustrates the computerized pairing operation controller **130** of FIG. **1**. The computerized pairing operation controller **130** includes a computerized processing device **250**, a communications device **260**, an input/output coordination device **270**, and a memory storage device **280**. It is noted that the computerized pairing operation controller **130** may include other components and some of the components are not present in some embodiments.

(31) The processing device **250** may include memory, e.g., read-only memory (ROM) and random-access memory (RAM), storing processor-executable instructions and one or more processors that execute the processor-executable instructions. In embodiments where the processing device **250** includes two or more processors, the processors may operate in a parallel or distributed manner. The processing device **250** may execute the operating system of the computerized pairing operation controller **130**. Processing device **250** may include one or more modules executing programmed code or computerized processes or methods including executable steps. Illustrated modules may include a single physical device or functionality spanning multiple physical devices. The processing device **250** may further include programming modules, including a trusted device identity module **252**, a vehicle onboarding module **254**, and a Bluetooth API management module **256**.

(32) The communications device **260** may include a communications/data connection with a bus device configured to transfer data to different components of the system and may include one or more wireless transceivers for performing wireless communication.

(33) The input/output coordination device **270** includes hardware and/or software configured to enable the processing device **250** to receive and/or exchange data with on-board resources of the vehicle **120**, including the touchscreen input/output device **134** of FIG. **1**, a microphone, a speaker, and buttons or other control inputs installed to the vehicle **120**.

(34) The memory storage device **280** is a device that stores data generated or received by the computerized pairing operation controller **130**. The memory storage device **280** may include, but is not limited to, a hard-drive or flash or solid-state memory.

(35) The trusted device identity module **252** includes programming to establish that a device to be paired with the vehicle **120** is a trusted device. The trusted device identity module **252** may include programming to utilize the validation certificate(s) to authenticate communications from the smartphone **110** of FIG. **1**. The trusted device identity module **252** may include stored lists of trusted devices.

(36) The vehicle onboarding module **254** includes programming to operate the disclosed system and method. The vehicle onboarding module **254** may include programming to initiate or control activation or operation of the notification signal useful to establish proximity of the vehicle **120** to the smartphone **110**. The vehicle onboarding module **254** may include programming automate establishing the classic Bluetooth connection with the smartphone **110**.

(37) The Bluetooth API management module **256** includes programming to enable communication through the established Bluetooth connection with the smartphone **110** and provide for a client-server relationship between the smartphone **110** and the vehicle **120**.

(38) The computerized pairing operation controller **130** is provided as an exemplary computerized device capable of executing programmed code to operate the disclosed system and method. A number of different embodiments of the computerized pairing operation controller **130** and

modules operable therein are envisioned, and the disclosure is not intended to be limited to examples provided herein.
(39) FIG. **4** is a flowchart illustrating a method **350** for vehicle Bluetooth pairing. The method **350** starts at step **352**. At step **354**, a vehicle onboarding application is installed to the vehicle **120** of FIG. **1**, and a smartphone onboarding application is installed to the smartphone **110** of FIG. **1**. Additionally, an authentication certificate may be provided to the smartphone **110**, and a validation certificate may be provided to the vehicle **120**. At step **356**, the vehicle **120** emits a notification signal, and the smartphone **110** monitors the notification signal. The smartphone **110** establishes proximity of the vehicle **120** to the smartphone **110** based upon the monitored notification signal. At step **358**, the vehicle **120** utilizes an automated pairing process to establish a classic Bluetooth connection with the smartphone **110**. At step **360**, the smartphone **110** and the vehicle **120** are connected through a Bluetooth connection and may transfer data securely therebetween. At step **362**, the method **350** ends. The method **350** is exemplary. A number of additional and/or alternative method steps are envisioned, and the disclosure is not intended to be limited to the examples provided.
(40) FIG. **5** is a flowchart illustrating a method **300** for initiating a vehicle Bluetooth pairing operation. The method **300** starts at step **302**. At step **304**, a notification signal or beacon is utilized to establish proximity of the vehicle **120** of FIG. **1** to the smartphone **110** of FIG. **1**. At step **306**, once proximity is established, the smartphone **110** is established as a trusted device through use of an authentication certificate in the smartphone **110** and through use of a validation certificate in the vehicle **120**. At step **308**, with proximity and with the smartphone **110** established as a trusted device, the pairing operation may be initiated. At step **310**, the method **300** ends. The method **300** is exemplary. A number of additional and/or alternative method steps are envisioned, and the disclosure is not intended to be limited to the examples provided.
(41) FIG. **6** is a dataflow **400** illustrating communication between a user **410**, an onboarding application **420**, and a remote computer resource **430** for a purpose of establishing simplified Bluetooth pairing. The user **410** represents the actions of a customer/user input to and output to the customer/user from an exemplary input/output device such as the input/output screen **112** of FIG. **1** and/or the touchscreen input/output device **134** of FIG. **1**. The onboarding application **420** represents computerized operations controlled by and information provided to the smartphone onboarding application of the smartphone **110** of FIG. **1** and/or the vehicle onboarding application of the vehicle **120** of FIG. **1**. The remote computer resource **430** represents computerized operations controlled by and information provided to the remote server device **140** of FIG. **1**. In operation **441**, the user **410** initiates or enables simplified vehicle Bluetooth pairing. In operation **442**, the onboarding operation provides information to the remote computer resource **430** to authenticate an identity of the user **410**. In operation **443**, the onboarding application **420** requests a vehicle identification number (VIN) for the vehicle **120** from the remote computer resource **430**. In operation **444**, the remote computer resource **430** provides the VIN to the onboarding application **420**. In operation **445**, the onboarding application **420** derives a beacon identifier. The beacon identifier may, in one embodiment, be described as a first function times the VIN. In operation **446**, the onboarding application **420** derives a BLE universally unique identifier (UUID), which may be described as a second function times the VIN. Exemplary Bluetooth identifier derivation functions include secure hash algorithm 128 (SHA-128), message-digest algorithm 5 (MD5), message-digest algorithm 4 (MD4), message-digest algorithm 2 (MD2), or other similar algorithms. In operation **447**, the onboarding application **420** registers the beacon identification. In operation **448**, the onboarding application **420** certifies to the user **410** that simplified vehicle Bluetooth pairing is enabled. The dataflow **400** is provided as an example. A number of additional and/or alternative operations are envisioned, and the disclosure is not intended to be limited to the examples provided.
(42) FIG. **7** is a dataflow **500** illustrating communication between a user **510**, a smartphone

onboarding application **520**, and a vehicle **530** for Bluetooth pairing process. The user **510** represents the actions of a customer/user input to and output to the customer/user from an exemplary input/output device such as the input/output screen **112** of FIG. **1** and/or the touchscreen input/output device **134** of FIG. **1**. The smartphone onboarding application **520** represents computerized operations controlled by and information provided to the smartphone onboarding application of the smartphone **110** of FIG. **1**. The vehicle **530** represents computerized operations controlled by and information provided to the vehicle **120** and/or the computerized pairing operation controller **130** of FIG. **1**. In operation **541**, the notification signal or beacon is found by the smartphone onboarding application **520**. In operation **542**, the smartphone onboarding application **520** prompts the user **510** to start the pairing process by providing an input. In operation **543**, the user **510** provides an input to start the pairing process. In operation **544**, the vehicle **120** establishes a secure BLE connection with the smartphone **110**. In operation **545**, identity verification is performed through use of the authentication certificate and the verification certificate. In operation **546**, the smartphone onboarding application **520** requests from the vehicle **530** a Bluetooth name. In operation **547**, the vehicle **530** provides to the smartphone onboarding application **520** the Bluetooth name. The Bluetooth name and other information transmitted between the entities will be signed with a certificate in order to establish and keep secure messaging between the entities. In operation **548**, the smartphone onboarding application **520** initiates an automated pairing process. In operation **549**, the smartphone onboarding application **520** requests a pairing code. In operation **550**, the vehicle **530** responds with a pairing code. In operation **551**, the smartphone onboarding application **520** confirms the pairing code, for example, by providing a prompt to a user of the smartphone **110** to confirm that the pairing code is correctly displayed upon both the smartphone **110** and a display of the vehicle **120**. In operation **552**, the smartphone onboarding application **520** provides a pairing complete confirmation to the user **510**. The dataflow **500** is provided as an example. A number of additional and/or alternative operations are envisioned, and the disclosure is not intended to be limited to the examples provided.

(43) FIG. **8** is a flowchart illustrating method **600** for operating a simplified vehicle Bluetooth pairing algorithm. The method **600** starts at step **602**. At step **604**, a vehicle VIN for the vehicle **120** of FIG. **1** is retrieved. At step **606**, Bluetooth identifiers are derived. At step **608**, the smartphone **110** of FIG. **1** registers for vehicle beacon scanning. At step **610**, vehicle beacon proximity callback is performed. At step **612**, BLE is established and secured between the smartphone **110** and the vehicle **120**. At step **614**, the vehicle identity is confirmed. At step **616**, a determination is made whether the vehicle identity is verified. If the vehicle identity is verified, the method **600** advances to the step **620**. If the vehicle identity is not verified, the method **600** advances to the step **618**. At step **618**, a determination is made whether the customer has exited the range defined for proximity. If the customer has not exited the range defined for proximity, the method **600** advances to the step **620**. If the customer has exited the range defined for proximity, the method returns to the step **608**.

(44) At step **620**, a classic Bluetooth pairing process is initiated between the smartphone **110** and the vehicle **120**. At step **622**, a determination is made whether the pairing code is verified. If the pairing code is not verified, the method **600** returns to the step **618**. If the pairing code is verified, the method **600** advances to the step **624**, where a user is notified of pairing success. The method **600** ends at the step **626**. The method **600** is exemplary. A number of additional and/or alternative method steps are envisioned, and the disclosure is not intended to be limited to the examples provided herein.

(45) FIG. **9** is a flowchart illustrating a method **700** for a customer to pair a Bluetooth enabled device with a vehicle including Bluetooth communications. The method **700** starts at step **702**. According to the disclosed system and method, the Bluetooth enabled device automatically determines proximity of the vehicle to the device. The device and the vehicle automatically authenticate each other. The device requests and the vehicle provides a Bluetooth name. The device requests a pairing code from the vehicle, and the vehicle provides the pairing code to the device.

The aforementioned processes occur automatically and include no steps to be taken by the customer. At step **704**, the device presents the pairing code to the customer and requests that the customer confirm that the pairing code provided on the device matches a pairing code displayed upon a screen installed to the vehicle. At step **706**, the customer provides a single touch input to the Bluetooth enabled device confirming that the pairing code displayed upon the device matches the pairing code displayed by the vehicle. Upon the customer providing the single touch input to the screen, the pairing is completed. The method **700** ends at step **708**. The method **700** is exemplary and may vary depending upon the brand of the Bluetooth enabled device and communication protocols programmed there into. A number of additional and/or alternative method steps are envisioned, and the disclosure is not intended to be limited to the examples provided herein.
(46) While the best modes for carrying out the disclosure have been described in detail, those familiar with the art to which this disclosure relates will recognize various alternative designs and embodiments for practicing the disclosure within the scope of the appended claims.

Claims

1. A system for vehicle Bluetooth pairing, the system comprising: a smartphone configured for Bluetooth communication, wherein the smartphone includes an authentication certificate; a vehicle including: a computerized pairing operation controller configured for Bluetooth communication, the computerized pairing operation controller including a verification certificate, the computerized pairing operation controller including programming to: emit a notification signal detectable by the smartphone, wherein the notification signal includes the verification certificate; and receive the authentication certificate from the smartphone; wherein the smartphone includes programming to: establish proximity to the vehicle based upon the notification signal; verify identity of the vehicle based upon the authentication certificate and the verification certificate; and establish a Bluetooth low energy (BLE) connection to the vehicle based upon establishing the proximity; and wherein the computerizing pairing operation controller further includes programming to automatically set up a Bluetooth connection after the Bluetooth low energy connection is established; and wherein the smartphone further includes programming to re-establish proximity to the vehicle when the smartphone has exited a range for establishing the proximity to the vehicle.
2. The system of claim 1, wherein the smartphone further includes programming to utilize the authentication certificate to provide an identity to the vehicle; and wherein the programming to automatically set up the Bluetooth connection is performed after the identity is verified.
3. The system of claim 1, wherein the notification signal includes data configured as an identifier specific to the vehicle; and wherein the programming to establish proximity to the vehicle based upon the notification signal establishes proximity to the vehicle based upon the data configured as the identifier specific to the vehicle.
4. The system of claim 3, wherein the data configured as an identifier specific to the vehicle is based upon a vehicle identification number for the vehicle.
5. The system of claim 1, wherein the programming to automatically set up the Bluetooth connection includes: the smartphone and the vehicle verifying identities of each other based upon the authentication certificate saved in the smartphone and the verification certificate saved in the vehicle; the smartphone requesting a Bluetooth name from the vehicle; the vehicle providing the Bluetooth name to the smartphone; the smartphone requesting a pairing code from the vehicle; and the vehicle providing the pairing code to the smartphone.
6. The system of claim 5, wherein the programming to automatically set up the Bluetooth connection further includes: the pairing code being displayed upon a display of the vehicle; the pairing code being displayed upon the smartphone; and completing the Bluetooth connection based upon a user providing confirmation of matching pairing codes.
7. The system of claim 1, wherein the programming to establish the Bluetooth low energy

connection to the vehicle includes deriving a Bluetooth low energy service identifier based upon a vehicle identification number for the vehicle.
8. The system of claim 1, wherein the programming to automatically set up the Bluetooth connection includes: monitoring a single touch input by a user to the smartphone; and completing the Bluetooth connection based upon the single touch input.
9. A system for vehicle Bluetooth pairing, the system comprising: a smartphone configured for Bluetooth communication, wherein the smartphone includes an authentication certificate; a vehicle including: a computerized pairing operation controller configured for Bluetooth communication, the computerized pairing operation controller including a verification certificate, the computerized pairing operation controller including a vehicle onboarding application including programming to: emit a notification signal detectable by the smartphone, wherein the notification signal includes the verification certificate; and receive the authentication certificate from the smartphone; wherein the smartphone includes a smartphone onboarding application including programming to: establish proximity to the vehicle based upon the notification signal; verify identity of the vehicle based upon the authentication certificate and the verification certificate; and establish a Bluetooth low energy connection to the vehicle based upon establishing the proximity; and wherein the vehicle onboarding application further includes programming to automatically set up a Bluetooth connection after the Bluetooth low energy connection is established; and wherein the smartphone further includes programming to re-establish proximity to the vehicle when the smartphone has exited a range for establishing the proximity to the vehicle.
10. The system of claim 9, wherein the programming to automatically set up the Bluetooth connection includes: the smartphone and the vehicle verifying identities of each other based upon the authentication certificate saved in the smartphone and the verification certificate saved in the vehicle; the smartphone requesting a Bluetooth name from the vehicle; the vehicle providing the Bluetooth name to the smartphone; the smartphone requesting a pairing code from the vehicle; and the vehicle providing the pairing code to the smartphone.
11. The system of claim 9, wherein the vehicle onboarding application and the smartphone onboarding application are configured for completing the Bluetooth connection based upon a single touch input by a user to the smartphone.
12. A method for vehicle Bluetooth pairing, the method comprising: within a first computerized processor of a vehicle, activating Bluetooth signal including a notification signal visible to Bluetooth enabled devices, wherein the notification signal includes a verification certificate; within a second computerized processor of a smartphone, wherein the smartphone includes an authentication certificate, receiving the notification signal; establishing proximity to the vehicle based upon the notification signal; verifying identity of the vehicle based upon the authentication certificate and the verification certificate; and establishing a Bluetooth low energy connection to the vehicle based upon establishing the proximity; and within the first computerized processor, automatically setting up a Bluetooth connection after the Bluetooth low energy connection is established; and re-establishing proximity to the vehicle when the smartphone has exited a range for establishing the proximity to the vehicle.
13. The method of claim 12, further comprising: within the second computerized processor, utilizing the authentication certificate to provide an identity to the vehicle; and within the first computerized processor, utilizing the verification certificate to verify the identity of the smartphone; and wherein automatically setting up the Bluetooth connection is performed after the identity is verified.
14. The method of claim 12, wherein the notification signal includes data configured as an identifier specific to the vehicle; and wherein establishing proximity to the vehicle based upon the notification signal includes establishing the proximity to the vehicle based upon the data configured as the identifier specific to the vehicle.
15. The method of claim 14, wherein the data configured as an identifier specific to the vehicle is

based upon a vehicle identification number for the vehicle.
16. The method of claim 12, wherein automatically setting up the Bluetooth connection includes: verifying identities of the smartphone and the vehicle based upon the authentication certificate saved in the smartphone and the verification certificate saved in the vehicle; the smartphone requesting a Bluetooth name from the vehicle; the vehicle providing the Bluetooth name to the smartphone; the smartphone requesting a pairing code from the vehicle; and the vehicle providing the pairing code to the smartphone.
17. The method of claim 16, wherein automatically setting up the Bluetooth connection further includes: the pairing code being displayed upon a display of the vehicle; the pairing code being displayed upon the smartphone; the smartphone prompting a user to provide confirmation of matching pairing codes; and completing the Bluetooth connection based upon the confirmation.
18. The method of claim 12, wherein establishing the Bluetooth low energy connection to the vehicle includes deriving a Bluetooth low energy service identifier based upon a vehicle identification number for the vehicle.
19. The method of claim 12, wherein automatically setting up the Bluetooth connection includes: monitoring a single touch input by a user to the smartphone; and completing the Bluetooth connection based upon the single touch input.